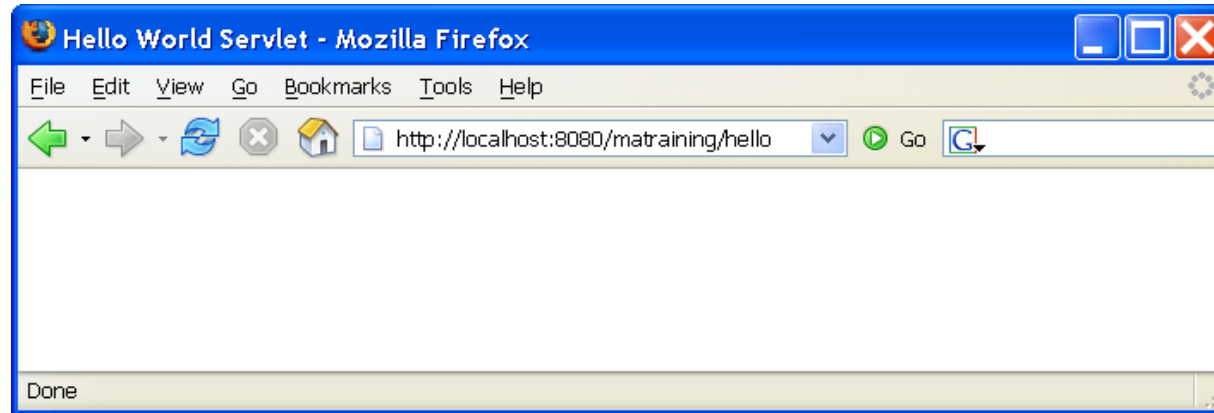


Web Applications in Java

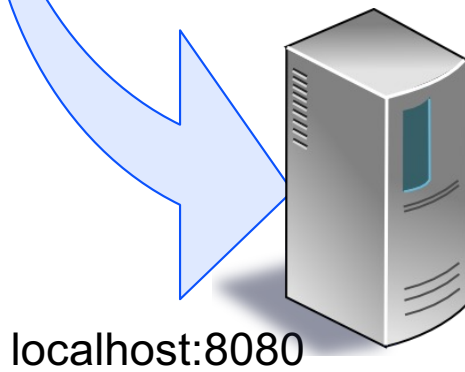
What's in a Java Web Application?

- Servlets
 - pre-compiled Java class files
- JavaServer Pages (JSPs)
 - HTML/XML pages with embedded Java scriptlets
- Supporting Java classes
 - Java Beans
 - Custom Tag libraries (TagLibs)
 - Filters
- Static Resources
 - HTML pages & CSS style sheets
 - XML & XSLT documents
 - Tag Library Descriptor files (TLD)
- (Optionally) A Deployment Descriptor (DD)
 - the `web.xml` defines web application environment

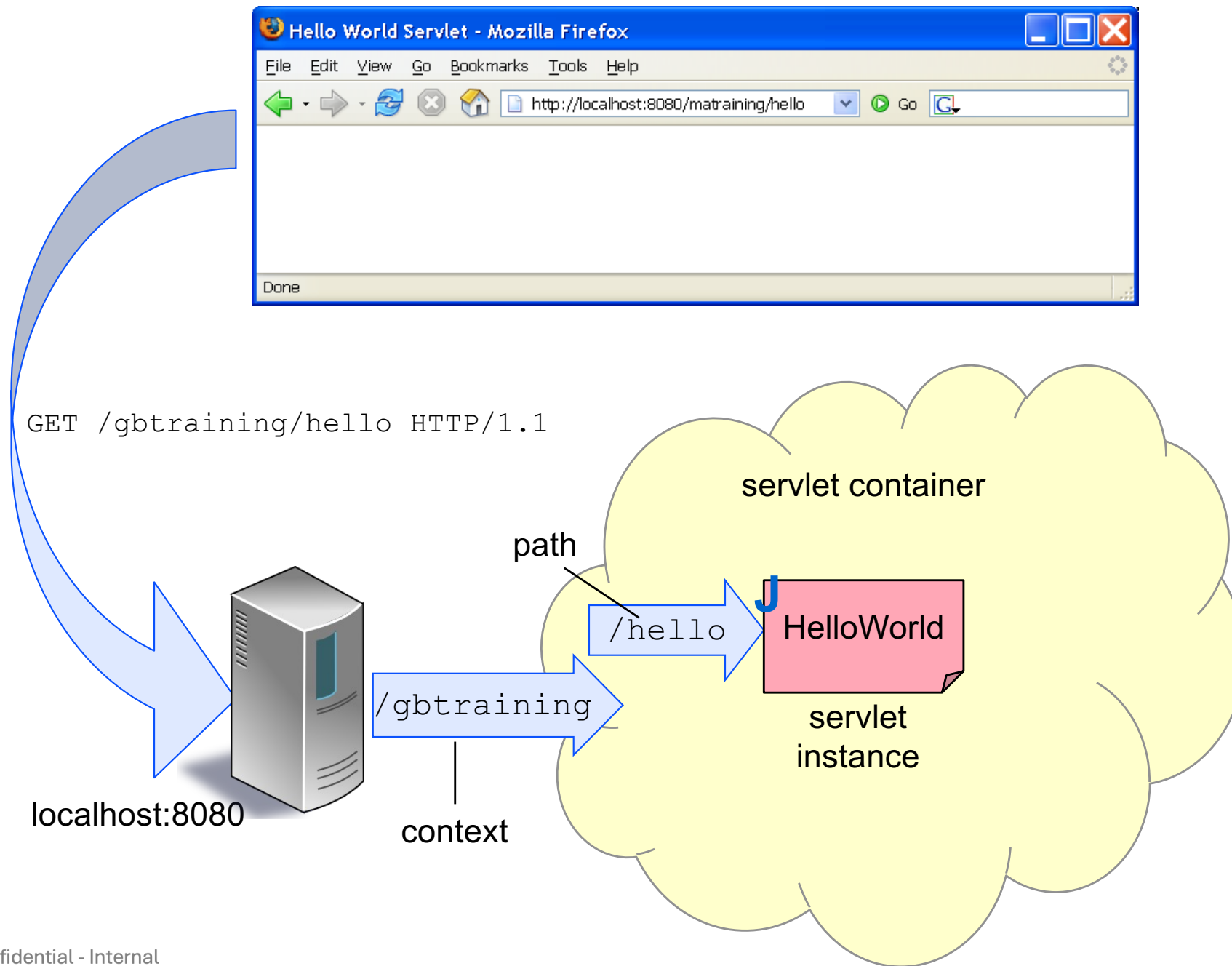
Servlets in Action



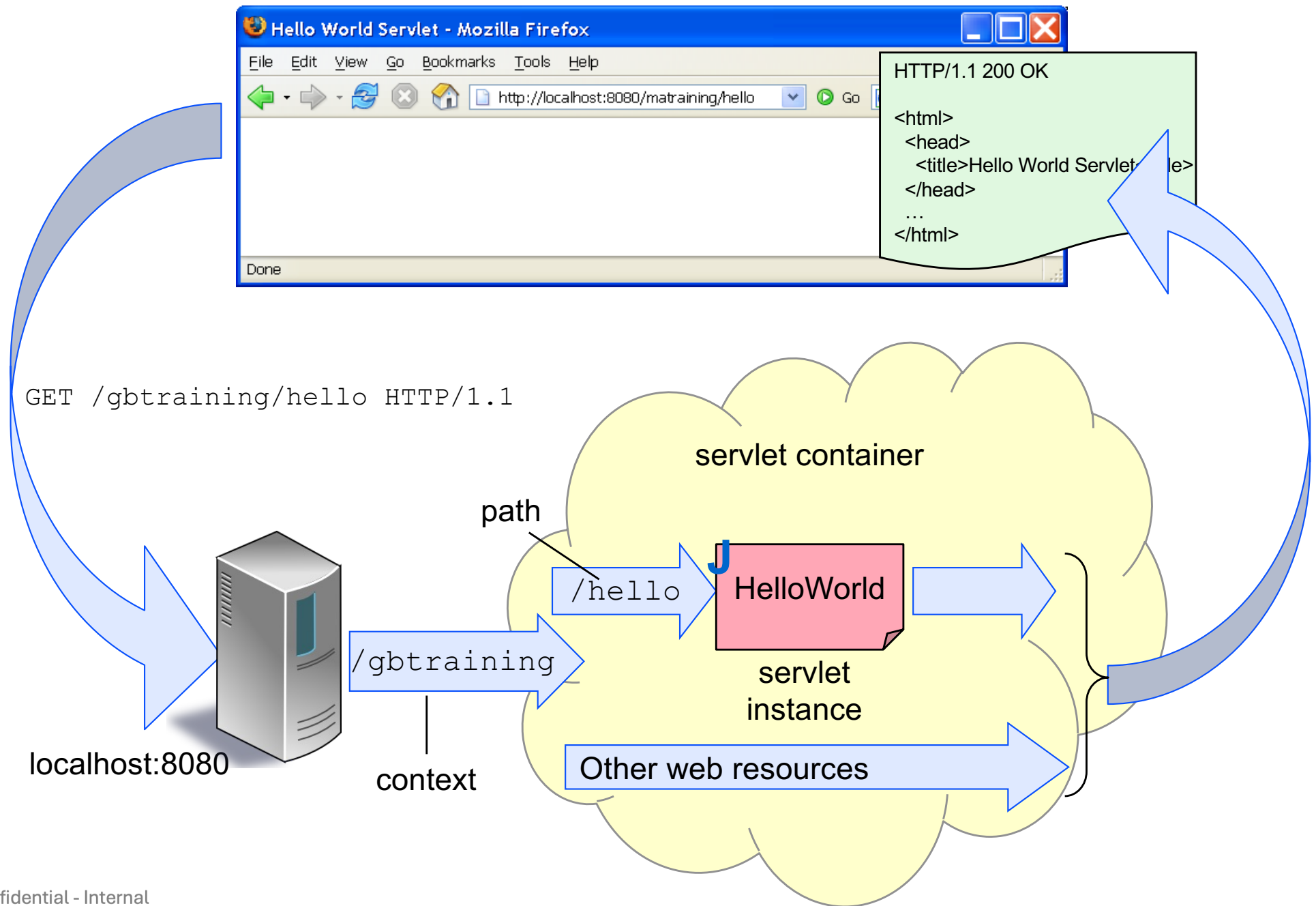
GET /gbtraining/hello HTTP/1.1



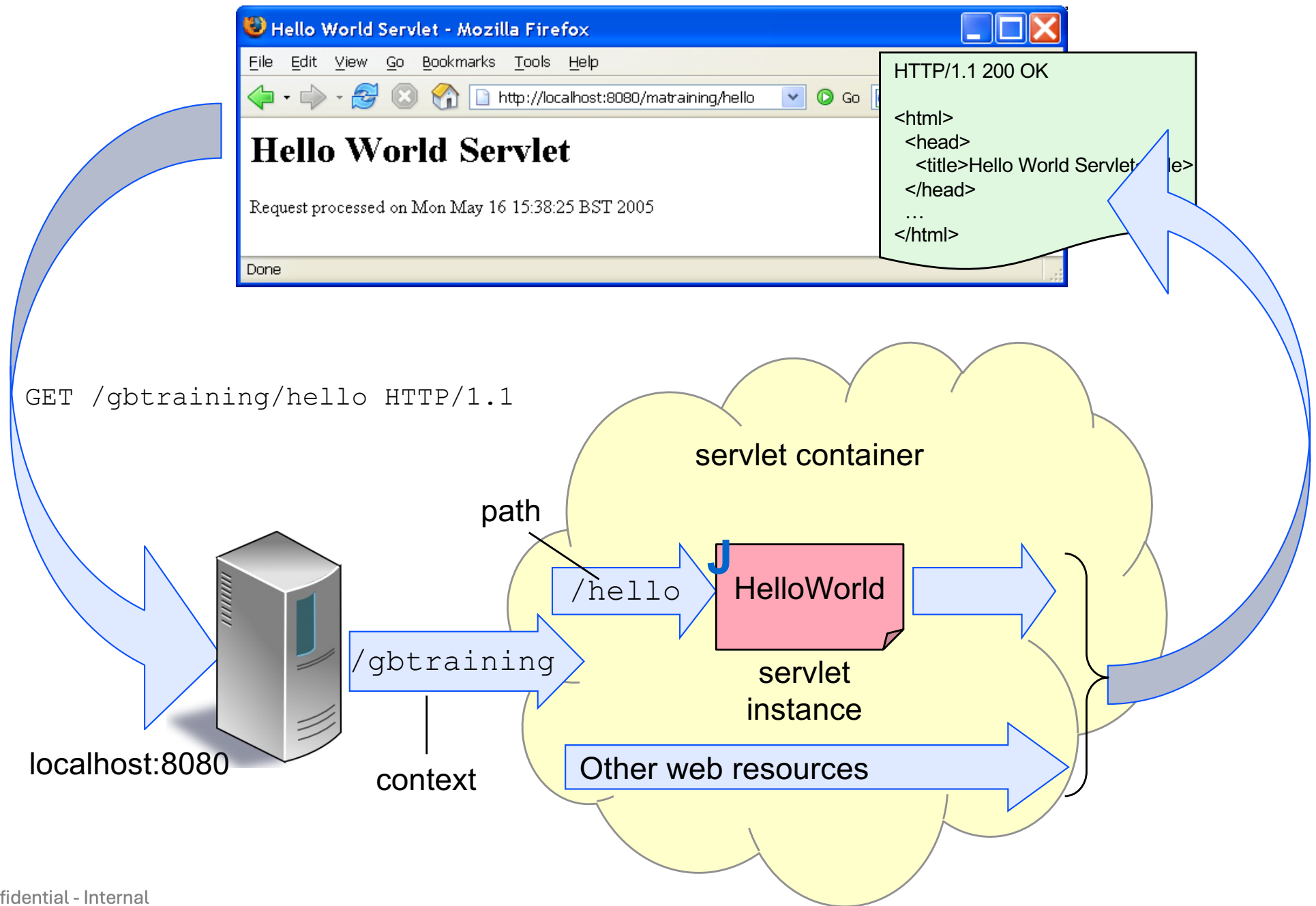
Servlets in Action



Servlets in Action



Servlets in Action



A Simple Servlet

```
package com.gbtraining.web;

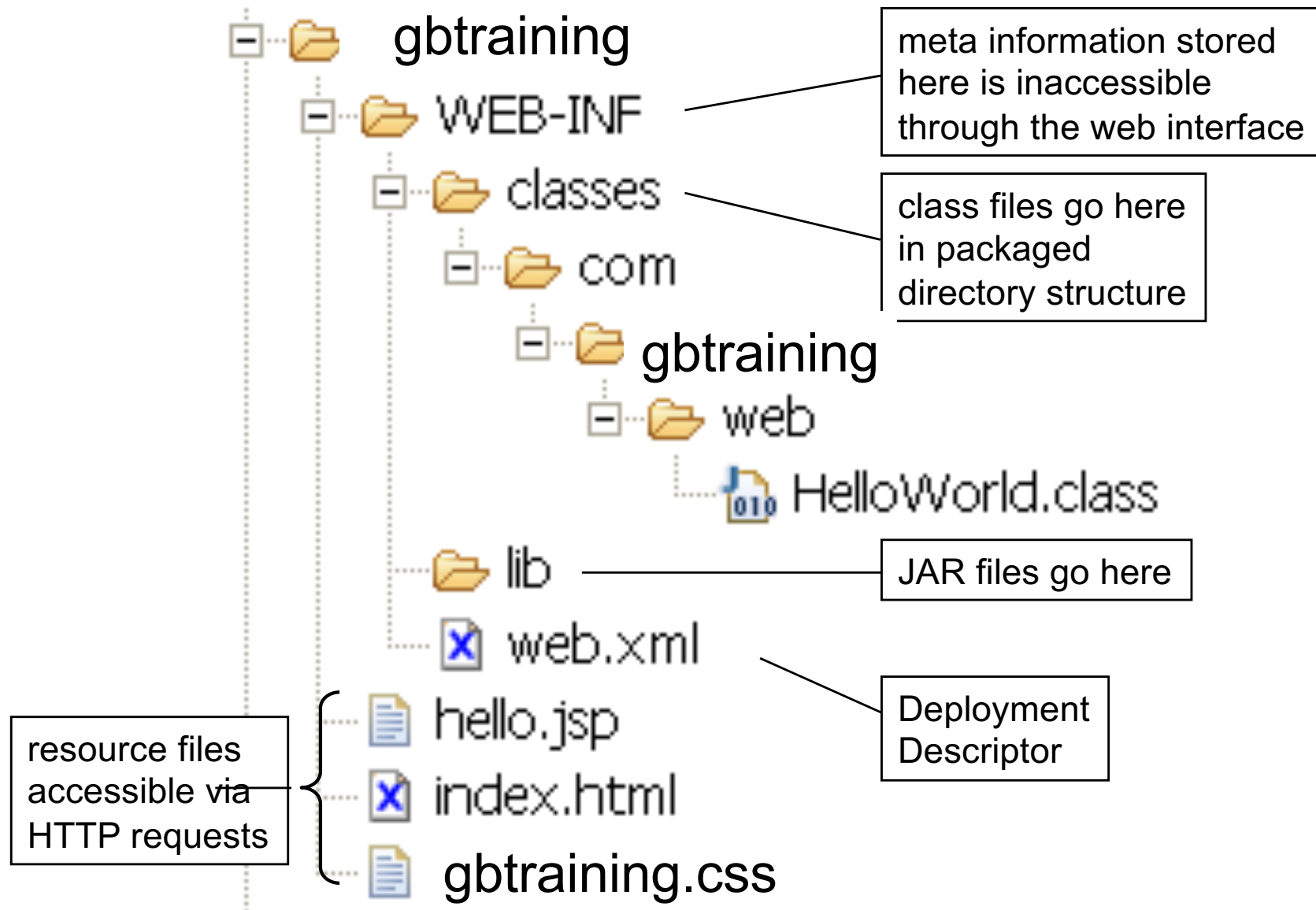
import java.io.*;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class HelloWorld extends HttpServlet {
    public void doGet( HttpServletRequest request,
                      HttpServletResponse response )
                      throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<HTML>");
        out.println("<HEAD><TITLE>Hello World Servlet</TITLE>");
        out.println("</HEAD><BODY>");
        out.println("<H1>Hello World Servlet</H1>");
        out.println("Request processed on " + new Date());
        out.println("</BODY></HTML>");
    }
}
```

More Servlet Methods

```
// Called by servlet container to initialize the servlet,  
// use in lieu of a constructor  
public void init() throws ServletException {  
    }  
  
// Called by servlet container when servlet instance  
// taken out of service  
public void destroy() {  
    }  
  
// Called for HTTP POST requests  
public void doPost( HttpServletRequest request,  
                   HttpServletResponse response )  
                   throws ServletException, IOException {  
    // Normally delegate processing to doGet() method  
    doGet(request,response);  
}
```


Java Web Application Structure



web.xml Deployment Descriptor

```
<?xml version="1.0" encoding="UTF-8"?>

<web-app xmlns="http://java.sun.com/xml/ns/j2ee"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://java.sun.com/xml/ns/j2ee
    http://java.sun.com/xml/ns/j2ee/web-app_2_4.xsd"
  version="2.4">

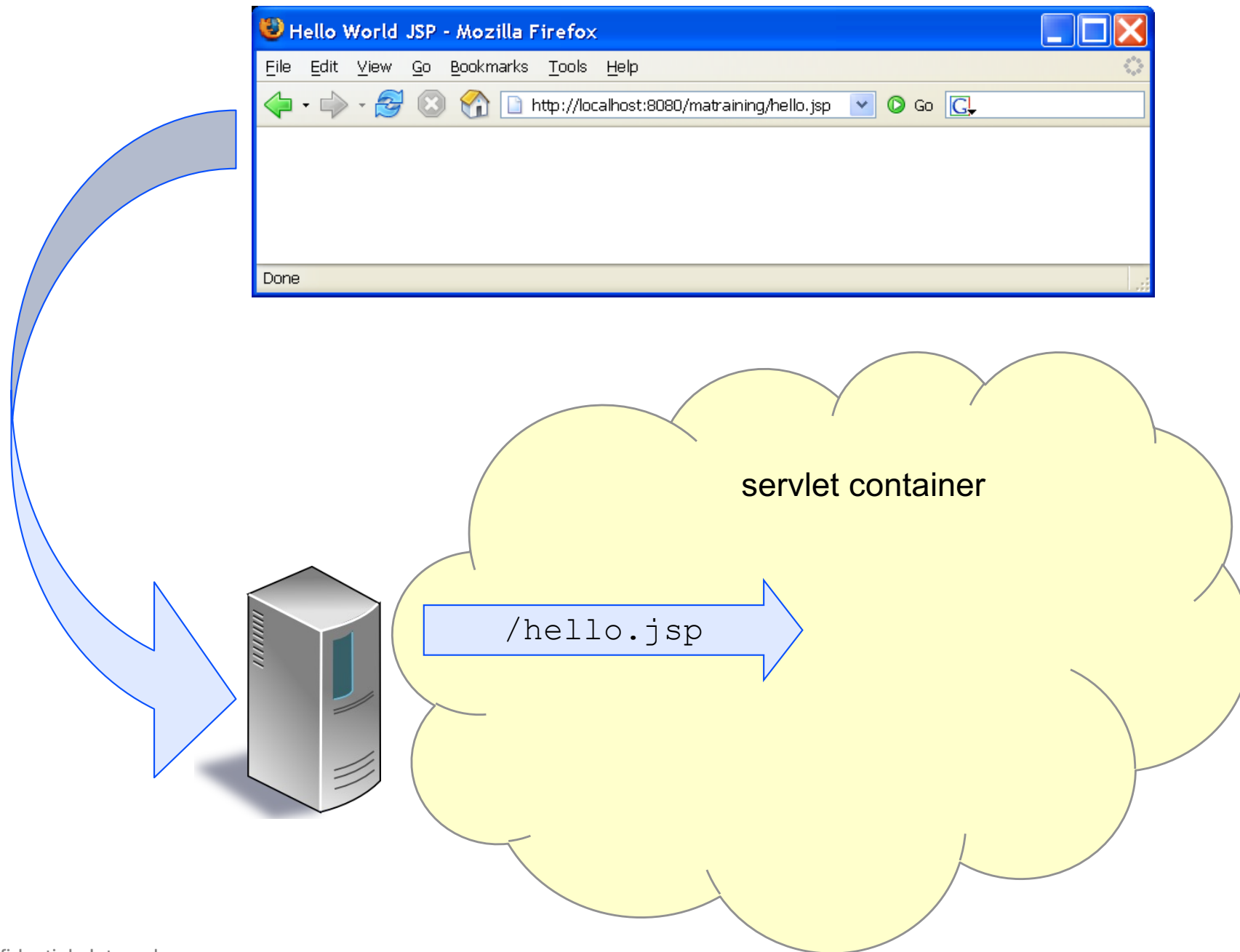
  <display-name>MA Training</display-name>

  <servlet>
    <servlet-name>HelloWorld</servlet-name>
    <servlet-class>com.maaining.web.HelloWorld</servlet-class>
  </servlet>

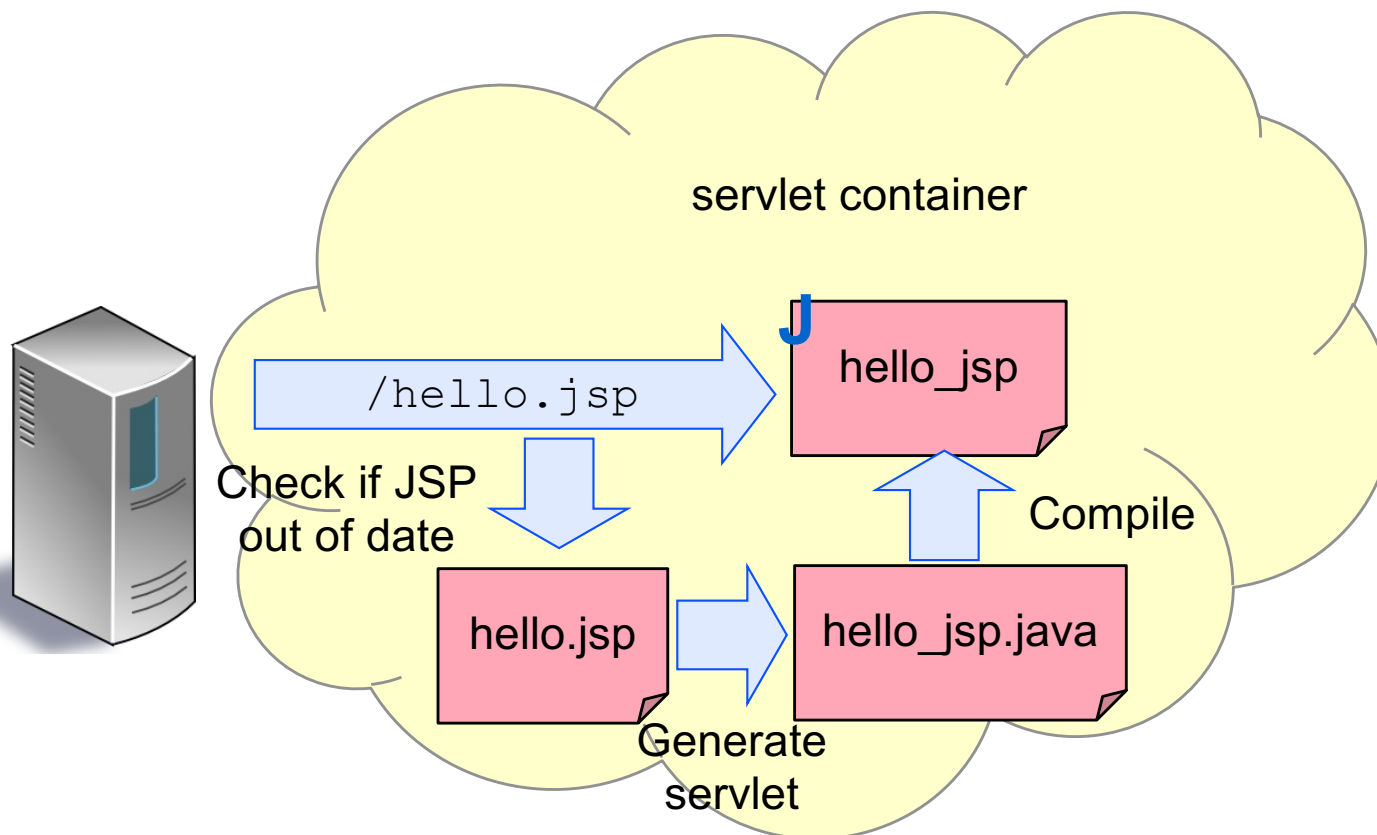
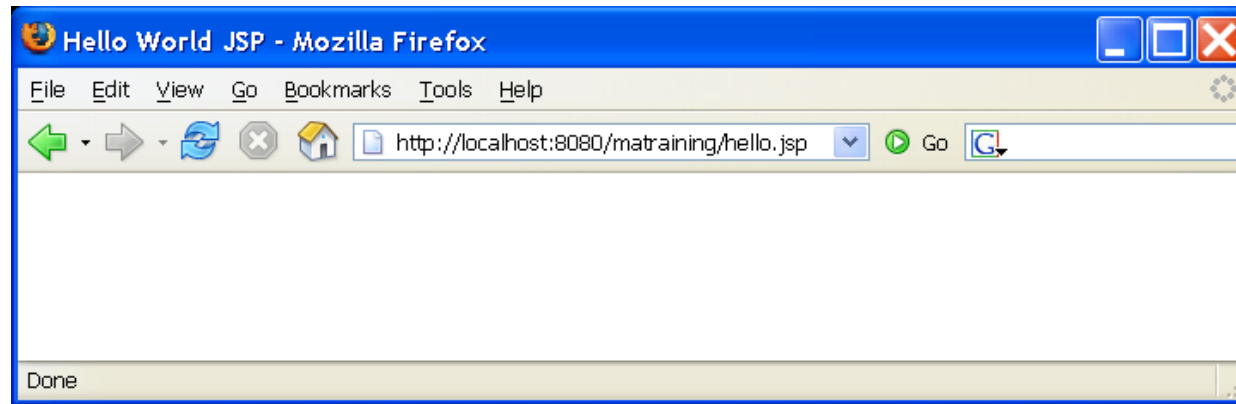
  <servlet-mapping>
    <servlet-name>HelloWorld</servlet-name>
    <url-pattern>/hello</url-pattern>
  </servlet-mapping>

</web-app>
```

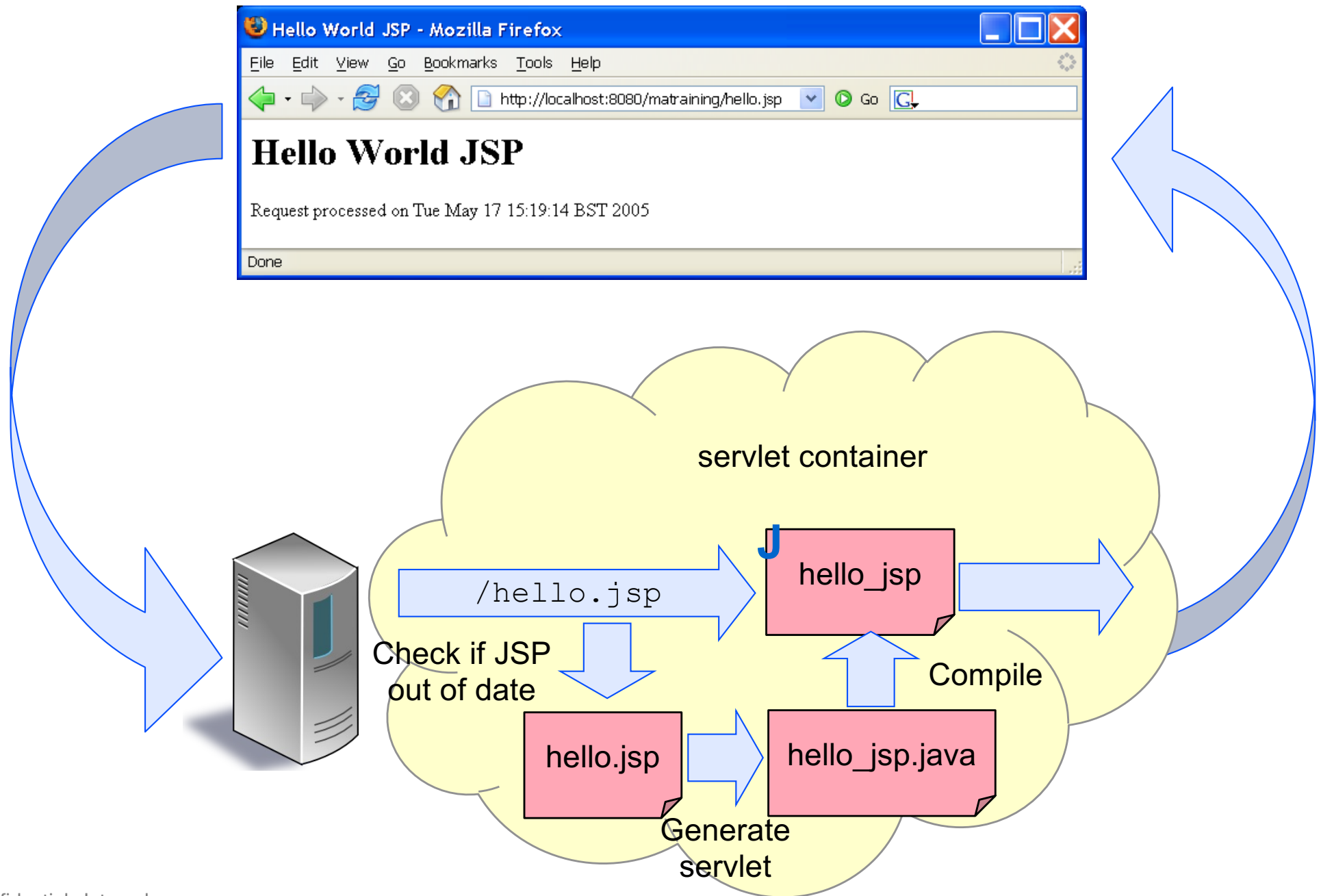
JSP in Action



JSP in Action



JSP in Action



A Simple JSP

```
<html>
<head>
  <title>Hello World JSP</title>
</head>
<body>
  <h1>Hello World JSP</h1>
  Request processed on <%= new java.util.Date()%>
</body>
</html>
```



embedded Java expression

Servlets and Threading

- Servlets are multi-threaded
 - use synchronized access shared mutable data

